Shengzhou **QIANG**

Ph.D. Candidate in Computer Science
Department of Computer Science and Engineering
University at Buffalo, The State University of New York

sqiang@buffalo.edu
sqiang.us
github.com/Alephant6
linkedin.com/in/sqiang

Education

2023– **Ph.D. Candidate in Computer Science** — **University at Buffalo**, Buffalo, NY, USA
Advisor: Prof. Haonan Lu
Advanced to candidacy, 2026
Research topics: distributed transactions, consistency protocols, evaluation of distributed datastores

2019–23 **B.S. in Information and Computing Science** — **Nanjing Univ. of Information Sci. & Tech.**, Nanjing, China
GPA: 3.8/4.0 (top 5%)
Industry-embedded program with Neusoft Group
First-class Outstanding Graduate of Jiangsu Province

2022 **Visiting student**, AI and information technology — **Macau Univ. of Science and Technology**, Macau, China
Funded by the Jiangsu Provincial Government Overseas Study Scholarship

Appointments

2023– **Graduate Research Assistant** — Advisor: Prof. Haonan Lu, University at Buffalo

2023–25 **Teaching Assistant** — Dept. of Computer Science and Engineering, University at Buffalo

2020–23 **Software Technology Trainee** — Neusoft Group, China

Research

2025– **SONIC** — read-only transactions without replication-critical designs *first author, under review*
- Proved that "replication-critical" read-only transaction designs are unnecessary: formalized the SONIC and Remote-Read theorems and demonstrated them in two purpose-built systems.
- Built **Starry** by deep-modifying the University of Washington's TAPIR codebase (C++) and extended COPS-SNOW with chain replication.
- Validated consistency with self-built Jepsen falsifier suites (Clojure); contributed fixes upstream to Jepsen.

2025– **Juicer** — transparent request-scheduling layer for low tail latency *co-author, in submission*
- Cut P99 latency by up to 75% and abort rate by up to 85% for strictly serializable databases, with no changes to database code, by reordering in-flight transaction messages by timestamp in a gRPC-Go interceptor.
- Outperformed NCC (SOSP'23) by 54% on tail latency, evaluated on CloudLab (9 × 16-core nodes, 25 Gbps).
- Implemented concurrency-control protocols (2PL wound-wait/wait-die, MVTO) in a Go distributed transactional key-value testbed; built its workload generator (Zipfian, dynamic skew, read-modify-write).

2025– **PRC** — developer-intended consistency testing for distributed datastores *co-author, in submission*
- Directed implementation testing that turns community-known correctness pitfalls (e.g., timestamp inversion) into targeted tests, complementing random-exploration tools such as Jepsen; evaluated the P formal-modeling framework and Fest.

2023–25 **EaaS** — Evaluation-as-a-Service for distributed storage systems
- Co-built the first end-to-end evaluation framework for distributed storage (C++, gRPC, Raft-replicated control plane); top contributor with approximately half of the team's 1,800+ commits over 20 months.
- Refuted EPaxos's published 3× throughput claim with cross-region AWS measurements (1–1.5× measured); uncovered an EPaxos livelock at 9 replicas; motivated a proven f·T upper bound on replica-utilization gains.

- Designed the unified workload engine (YCSB, transactional YCSB-T, interactive TPC-C) with C++/Go/Java adapters integrating CockroachDB, Cassandra, and EPaxos-family protocols at approximately 20 lines of code per system.

Publications

Manuscripts under review

2026 **Qiang, S.** et al. SONIC: rethinking replication-critical read-only transactions. *Under review at a top-tier systems conference.* (first author)

2026 Juicer: transparent request scheduling for low tail latency. *In submission.* (co-author)

2026 PRC: developer-intended consistency testing for distributed datastores. *In submission.* (co-author)

2025 Enabling easy, correct, and reproducible evaluation of distributed storage systems with Evaluation as a Service. *OSDI submission.* (co-author)

Journal articles

2021 **Qiang, S.-Z.**, Jiang, X., Han, P.-Y., et al. (2021). Instantaneous cross-correlation function type of WD based LFM signals analysis via output SNR inequality modeling. *EURASIP Journal on Advances in Signal Processing*, 2021, 122. doi:10.1186/s13634-021-00830-7 (first author)

2022 Shi, X.-Y., Wu, A.-Y., Sun, Y., & **Qiang, S.-Z.** (2022). Unique parameters selection strategy of linear canonical Wigner distribution via multiobjective optimization modeling. *Chinese Journal of Electronics.* doi:10.23919/cje.2021.00.338

2022 Wu, A.-Y., et al. (2022). A computationally efficient optimal Wigner distribution in LCT domains for detecting noisy LFM signals. *Mathematical Problems in Engineering*, 2022, 2036285. doi:10.1155/2022/2036285

2022 Liu, T., Yin, M.-T., Rong, J.-P., **Qiang, S.-Z.**, et al. (2022). Modulation instabilities in twin-core fibers with self-steepening effects. *Physics Letters A*, 425, 127869.

2021 Zhang, Z.-C., **Qiang, S.-Z.**, Jiang, X., Han, P.-Y., Shi, X.-Y., & Wu, A.-Y. (2021). Linear canonical Wigner distribution of noisy LFM signals via variance-SNR based inequalities system analysis. *Optik*, 237, 166712. (second author)

2021 Zhang, Z.-C., Jiang, X., **Qiang, S.-Z.**, Sun, A., Liang, Z.-Y., Shi, X., & Wu, A.-Y. (2021). Scaled Wigner distribution using fractional instantaneous autocorrelation. *Optik*, 237, 166691. (third author)

2021 Zhang, Z.-C., Han, Y.-P., Sun, Y., Wu, A.-Y., Shi, X.-Y., **Qiang, S.-Z.**, et al. (2021). Heisenberg's uncertainty principle for N-dimensional fractional Fourier transform of complex-valued functions. *Optik*, 242.

Software copyrights

2021–22 Three registered software copyrights (one as sole author).

Teaching

2025 **CSE 521**: Operating Systems (Spring) graduate

2024 **CSE 586**: Distributed Systems (Fall) graduate

2023 **CSE 587**: Data-Intensive Computing (Fall) graduate

Awards and Scholarships

2024 **First-class Outstanding Graduate of Jiangsu Province**, Jiangsu Provincial Department of Education

2022 **Top 10%** (103/1079), Kaggle Foursquare Location Matching competition

2021 **Second Prize, Jiangsu Province**, China Undergraduate Mathematical Contest in Modeling

2021 **Second Prize, Jiangsu Province**, 12th "Beidou Cup" National Youth Science and Technology Innovation Contest

2021 **Jiangsu Provincial Government Overseas Study Scholarship**

2020–22 **First-class Academic Scholarship** and **Merit Student**, Nanjing Univ. of Information Sci. & Tech.

Research Projects (pre-Ph.D.)

2020–22 Key member, National Natural Science Foundation of China Youth Program — time-frequency distribution theory in linear canonical domains

2020 Principal investigator, National Undergraduate Innovation Program — spatio-temporal super-resolution in linear canonical domains

2021–22 Key member, National Undergraduate Innovation Program — modulation instability in nonlinear twin-core fibers

Skills

Languages C++, Go, Java, Python, Clojure, MATLAB, Shell, LaTeX

Systems Distributed transactions and concurrency control (2PL, OCC, MVCC-MVTO); replication and consensus (Paxos, Raft, EPaxos, chain replication, TAPIR); distributed key-value and storage internals (CockroachDB, Cassandra); gRPC

Evaluation YCSB / YCSB-T / TPC-C benchmarking, tail-latency analysis, Jepsen consistency testing, P formal modeling

Infra CloudLab, AWS multi-region, Docker Compose cluster orchestration, Linux, GitHub Actions, Proxmox

Big data Hadoop, Hive, Spark, Storm, HBase